EXECUTIVE LEADERSHIP GUIDE • V1.3 DESKTOP SUITE

Under 40 Report Dashboard

Contractor Weekly & Monthly Hours Reconciliation, Leave Audit & Automated Compliance Suite

TARGET AUDIENCE	CORE GOAL	ARCHITECTURE	DATA SECURITY
Managers, Dispatch & Payroll	Zero Shortfall Disputes & Invoicing Precision	6 Integrated Audit Tabs + Auto-Updater	100% Client-Side (Zero Cloud Leakage)

1. Executive Summary: Why This System Exists

In healthcare, logistics, and field service operations, full-time contractors are engaged to fulfill a **mandatory 40-hour work week**. However, candidate time is tracked across two completely disconnected software platforms every single week:

1. **Report 1 (Working Hours):** Timecard exports logging actual clock-in/clock-out shifts on the job.
2. **Report 2 (Leaves Details):** HR records logging medical appointments, statutory sick slips, and doctor notes.

Historically, operations managers had to spend **2 to 3 hours every Monday morning** manually opening both spreadsheets in Excel, running imperfect VLOOKUP formulas, reading freehand doctor comments, converting fractional days into hours, and reconciling candidate names. This manual drag led to frequent calculation mistakes, unfair pay deductions, and contentious client billing disputes.

The **Under 40 Report Dashboard** completely automates this entire audit. The user simply **drags and drops both weekly files** into the application. In less than **2 seconds**, the tool sanitizes the data, intelligently parses leave hours from freehand doctor notes, matches worker identities, and flags anyone whose adjusted total falls **strictly under 40 hours**.

< 2 Sec	100%	100+ Hours	Zero Cloud
RECONCILIATION TIME	CALCULATION PRECISION	SAVED PER YEAR	PRIVATE & OFFLINE READY

2. Key Problems Solved for Executive Leadership

⏱️ Eliminates Manual Excel Reconciliations

Replaces fragile formulas, copy-paste drudgery, and human spreadsheet errors with a standardized, one-click drag-and-drop workflow that runs instantaneously.

🩺 Parses Freehand Medical & Sick Notes

Intelligently interprets unstructured HR comments (e.g. "*took 4 hours leave*", "*2.5 hrs doctor*", or full days defaulting to 8h) without manual data-entry errors.

⚖️ Fast Operational Accountability

Equips dispatch and managers with instant dropdowns to assign accountability (Client Fault vs. Candidate Fault), set forms of pay, and track client notifications.

🔒 Enterprise-Grade Privacy & Data Security

Executes 100% locally with zero cloud exposure. Features full code obfuscation, inspection lockout (F12 and Right-Click disabled), anti-tampering, and complete RAM-only processing.

3. System Architecture & The 6-Tab Reference Matrix

The v1.3 compliance suite organizes contractor audit into **6 specialized operational tabs**, separating raw input data, daily managerial actions, audit mathematics, workforce rosters, and monthly executive consolidations:

Tab Name & Icon	Primary Question It Answers	Core Business Process Handled	Primary Stakeholders
📋 **Tab 1: Under 40 Report**	*"What is the exact mathematical shortfall for each non-compliant candidate?"*	Mathematical shortfall audit: Original Worked Hours + Leaves Hours + Statutory Holidays vs. 40-hour requirement.	Operations Auditors, Compliance Officers
🗂️ **Tab 2: All Processed**	*"What is the compliance health of the entire organization this week?"*	Master workforce roster: Side-by-side view with color-coded status badges (**Compliant** vs **Under 40h**).	Senior Management, HR Directors
⏱️ **Tab 3: Working Hours**	*"Where did the original worked hours numbers originate?"*	Raw timesheet transparency: Instant search, sorting, and row-level shift verification without opening Excel.	Timekeepers, Dispatchers
🩺 **Tab 4: Leaves Details**	*"Where did the sick leave and medical hours originate?"*	Medical audit trail: Inspect unedited doctor certificates, date ranges, and original employee leave notes.	HR Coordinators, Medical Auditors
🌟 **Tab 5: Grand Summary**	*"What are the cumulative candidate shortfalls across the entire month?"*	Monthly consolidation: Ingests 4–5 weekly audits, aggregates deficits, and transfers monthly data to the action sheet.	Billing Managers, Executive Leadership
📊 **Tab 6: Summary Report**	*"Who is at fault for the shortfall and how will each case be paid?"*	Dedicated operational action sheet: Log Client Fault, assign Form of Pay, manage Leave Balances, and write fresh notes.	Managers, Dispatch Leads, Payroll

4. End-to-End System Workflow (How Data Flows)

Phase 1: Ingestion & Sanitization
The dashboard reads both CSVs in parallel, normalizes irregular time strings (e.g. `24:00` → `24:00:00`), and removes test accounts and zero-hour placeholders.

Phase 2: Note Extraction & Fuzzy Matching
Regex algorithms scan freehand comments for partial hours (defaulting to 8h for full-day sick slips), reconciling candidate names across files via smart name matching.

Phase 3: Dual Compliance Triage
Candidates ≥ 40 hours are certified Compliant. Candidates < 40 hours are highlighted in red and populated into the mathematical audit and operational action sheets.

Phase 4: Decision Logging & White Paper Export
Managers assign fault, payment instructions, and fresh notes, then export clean white paper PDFs/CSVs via the interactive report chooser modal or save audit states.

> **Architectural Independence:** The **Summary Report** and **Grand Summary** operate as independent executive layers. Weekly timecard data remains pure and unpolluted, while managerial notes and monthly cumulative totals are tracked in dedicated action buffers.

5. Dedicated System Validations & Enterprise Security Checks

To guarantee absolute calculation integrity, prevent payroll disputes, and protect sensitive contractor medical data, the system implements **14 automated validation checks** and **10 enterprise security safeguards**:

Part A: Multi-Layer Validation Engine (14 Built-In Verification Checks)

📂 Layer 1: Ingestion & Schema Guards — 4 Checks

1. **Format Whitelist:** Strictly admits `.csv` files, or encrypted `.json` / `.u40` session state files.
2. **Non-Empty File Guard:** Rejects blank or corrupted uploads immediately to prevent silent downstream errors.
3. **Report 1 Schema Check:** Verifies mandatory *"Candidate Name"* and *"Working Hours"* columns before processing.
4. **Report 2 Schema Check:** Enforces mandatory *"Candidate Name"* and leave details columns before linking.

👥 Layer 2: Roster & Eligibility Filters — 5 Checks

5. **Blank Name Sanitizer:** Automatically detects and skips rows with null, whitespace, or missing candidate names.
6. **Test Account Blacklist:** Filters out test entries (e.g. *"TEST Peter"*, *"PENDING SHIFT"*) from the audit.
7. **Jobs Plus Role Filter:** Enforces candidate title: *"ni - employed and fully registered with jobs plus"*.
8. **Zero-Hours Exclusion:** Discards placeholder rows with 0 worked hours or blank clock-in times.
9. **Part-Timer Tracking:** Excludes contracted part-timers from the deficit list while keeping them restorable.

⚙️ Layer 3: Parsing & Smart Matching — 3 Checks

10. **Time Format Normalization:** Converts `HH:MM:SS`, `HH:MM`, and decimal hours into standard decimals.
11. **Smart Name Matching:** Reconciles candidate names across files despite middle initials or minor typos.
12. **Regex Comment & Date Parser:** Extracts partial hours (*"X hrs"*, *"X mins"*) with a safe 8h full-day fallback.

🎯 Layer 4: Mathematical Integrity — 2 Checks

13. **Strict 40.00h Threshold:** Gating logic: *Adjusted Hours < 40.00* tags shortfall; ≥ 40.00 tags compliant.
14. **Reactive Recalculation:** Validates overrides and recalculates weekly totals in real-time across all views.

Part B: Enterprise Security, Privacy & Integrity Protections (10 Core Safeguards)

🛡️ 1. Code Obfuscation & Logic Shielding

Proprietary calculation algorithms and business rules are compiled and scrambled into hexadecimal tokens, Base64 encrypted strings, and control-flow state machines.

🚫 2. Inspection & DevTools Lockout

Right-click inspection and developer shortcuts (`F12`, `Ctrl+Shift+I`, `Ctrl+U`) are blocked. The desktop executable enforces `--disable-dev-tools`.

🔒 3. Zero Cloud Leakage (100% Client-Side)

All CSV parsing, hours calculations, and report rendering execute in local device RAM. No contractor data is ever uploaded to external cloud servers.

🩺 4. Medical Data Privacy (GDPR / HIPAA)

Sensitive doctor notes, medical diagnoses, and sick certificate numbers remain strictly on the local workstation, safeguarding employee health privacy.

🌐 5. Air-Gapped & Offline Operational Ready

Operates flawlessly without an active internet connection. Perfect for secure corporate environments, VPNs, and air-gapped payroll stations.

📦 6. Sandboxed Desktop & Anti-Tamper

Runs as a standalone desktop executable (`Under40Report.exe`). Modified local files are automatically self-healed and restored from embedded resources on launch.

🧼 7. Input Sanitization & Anti-XSS

All parsed candidate names, doctor comments, and user overrides are strictly escaped prior to DOM rendering, preventing HTML/script injection attacks.

📑 8. Tamper-Evident Session Serialization

Saved audit files (`.json`/`.u40`) use deterministic schemas, allowing compliance officers to verify audit provenance and review overrides.

🔑 9. Zero Persistent Credential Storage

The software does not request, store, or cache employee bank details, social security numbers, or system passwords on disk.

🧹 10. Memory-Only Session Lifecycle

All active memory buffers are completely cleared when the application window is closed or when the user clicks *"Reset / Refresh All Data"*.

Executive Compliance Guarantee: The combination of 14 deterministic validation checks, binary code obfuscation, inspection lockout, and enterprise privacy safeguards makes this dashboard fully audit-proof, tamper-resistant, and compliant with corporate IT security standards.

6. Tab 1 Walkthrough: 📊 Summary Report (Manager Action Center)

The **Summary Report** is the primary operational workspace where dispatch leads and managers investigate shortfalls, attribute fault, assign payroll methods, and document operational explanations:

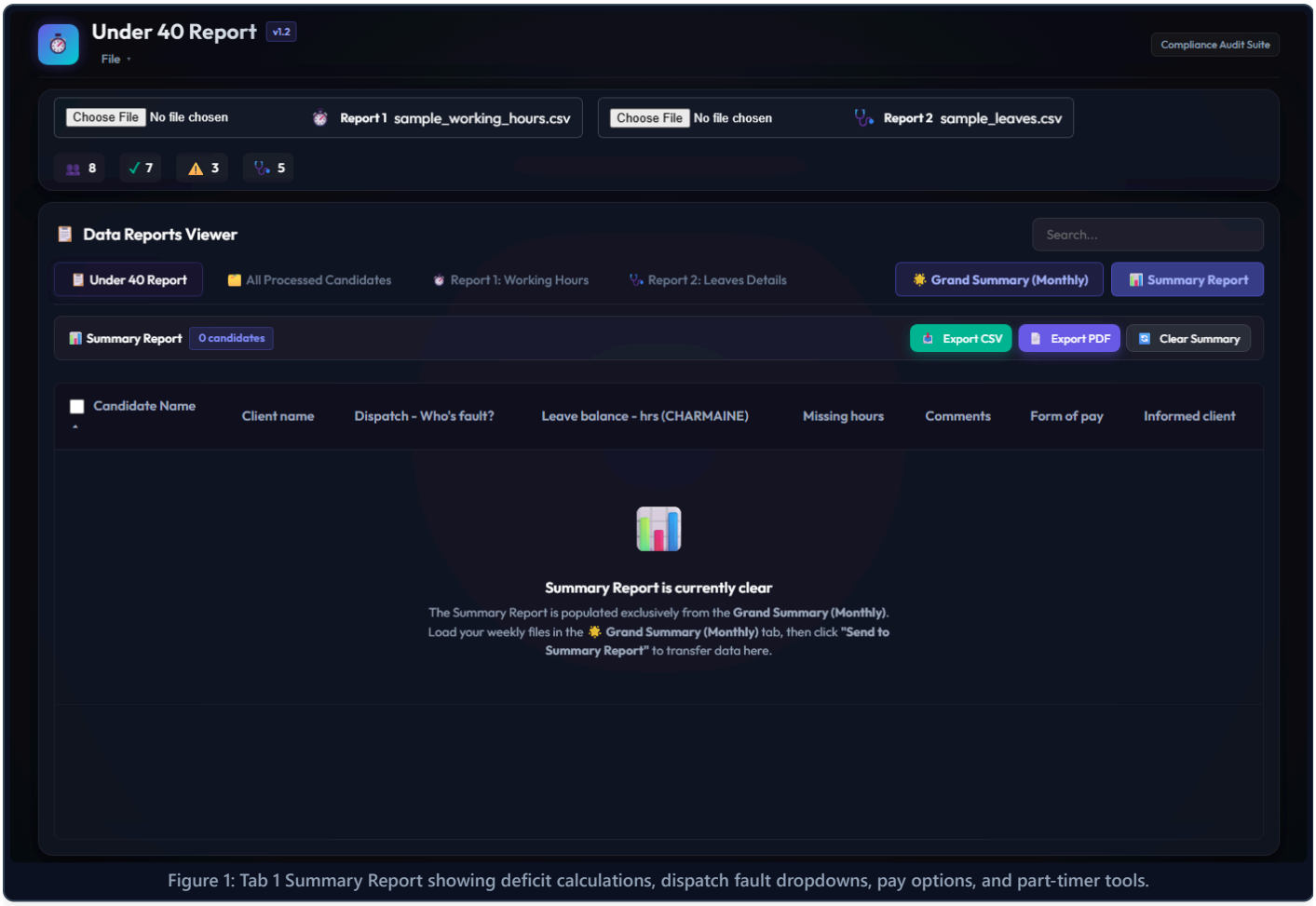


Figure 1: Tab 1 Summary Report showing deficit calculations, dispatch fault dropdowns, pay options, and part-timer tools.

Core Operational Fields in the Summary Report:

1. Missing Hours & Adjusted Total
Displays the exact shortfall in hours and minutes (e.g. *Missing: 01:00:00, Total: 39:00:00*). Zero manual math is required.

2. Dispatch - Who's Fault? Dropdown
Records accountability: **Client** (cancelled shift / site closure), **Candidate** (unexcused absence), or **Client+Candidate**.

3. Form of Pay Dropdown
Defines payroll action: **Paid Leave**, **Unpaid Leave**, **Out of Pocket**, or custom split pay codes directly per candidate row.

4. Leave Balance & Client Notification
Allows tracking accrued candidate leave balances in hours and logging whether the client has been formally notified (**Yes / No**).

Fresh Operational Comments vs. Medical Notes

Clean Operational Workspace: The Comments box in the Summary Report starts completely blank (it does not import doctor notes from Report 2). This gives dispatch and managers a clean, dedicated space to document fresh operational explanations regarding why shifts were missed.

Managing Part-Time Contractors (Batch Actions & Restore)

Not every worker under 40 hours is non-compliant; some are contracted for 20h or 30h per week. Managers can select checkboxes next to part-time workers and click "🗑️ **Delete Part-Timer(s)**". This cleanly removes them from the shortfall audit, while keeping them safely stored in a "↩️ **Part-Timers**" restore drawer in case they need to be re-added later.

Dedicated Summary Actions Bar

The Summary Report features its own dedicated action controls: **Export CSV**, **Export PDF** (clean white paper report), and **Clear Summary**. This ensures summary actions operate independently from weekly raw files.

7. Detailed Candidate Breakdown & Calculation Visualizer

Clicking on any candidate row in the Under 40 or Summary Report expands an interactive **Calculation Drawer**. This eliminates confusion by presenting an itemized, mathematically verifiable receipt of how weekly hours were determined:

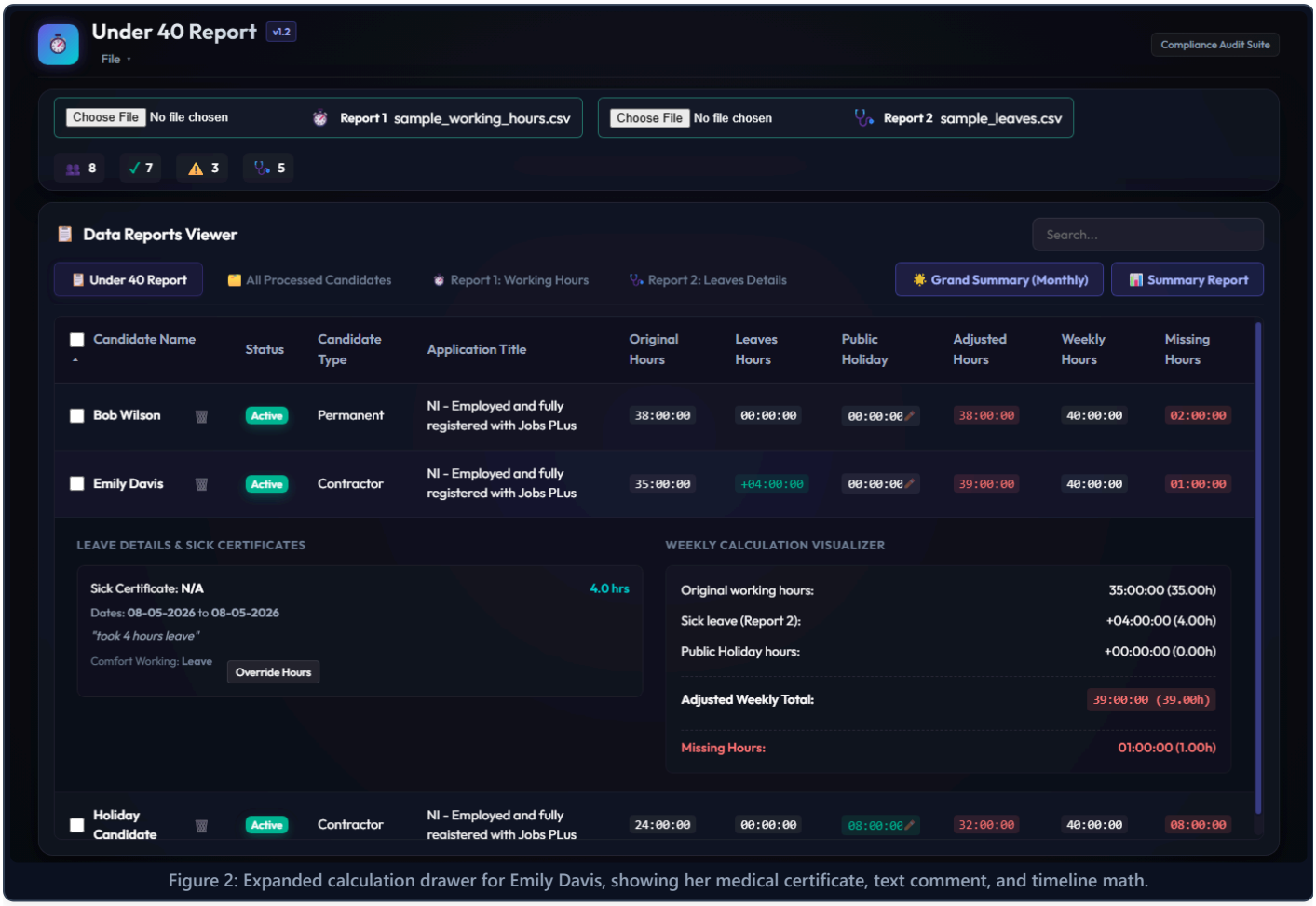


Figure 2: Expanded calculation drawer for Emily Davis, showing her medical certificate, text comment, and timeline math.

Real-World Case Study: Understanding Emily Davis

In messy real-world spreadsheets, candidate **Emily Davis** logged 35 hours of work and submitted a doctor slip with a freehand comment: "*took 4 hours leave*". Here is how the automated engine reconciled her week:

1. Logged Working Hours (Timecard Report 1):	35:00:00 (35.00 hrs)
2. Approved Leave Hours Extracted from Note ("took 4 hours leave"):	+04:00:00 (4.00 hrs)
3. Public Holiday Credit (Editable statutory override):	+00:00:00 (0.00 hrs)
4. Adjusted Weekly Total (Actual + Approved Leave):	39:00:00 (39.00 hrs)
5. Final Audit Flag: Short of 40h standard requirement by:	**01:00:00 (Missing 1 Hour)**

> **Why This Protects the Organization:** In manual Excel workflows, someone skimming the report might miss the freehand note "*took 4 hours leave*" and dock Emily 5 hours of pay. The tool extracted the note, credited her 4 hours, and correctly identified that she is only 1 hour short—preventing contractor payroll disputes before they happen.

Interactive Leave Overrides & Holiday Credits

If management approves an exceptional adjustment, managers can click **"Override Hours"** directly inside the drawer or enter **Public Holiday** credits. The adjusted total and missing hours recalculate across the entire dashboard instantaneously.

8. Tab 2 Walkthrough: 📋 Under 40 Report (Mathematical Audit Engine)

While the Summary Report is built for operational actions, the **Under 40 Report** is designed for compliance auditors and payroll accountants. It displays the clean mathematical equation for every candidate who failed to reach 40 hours:

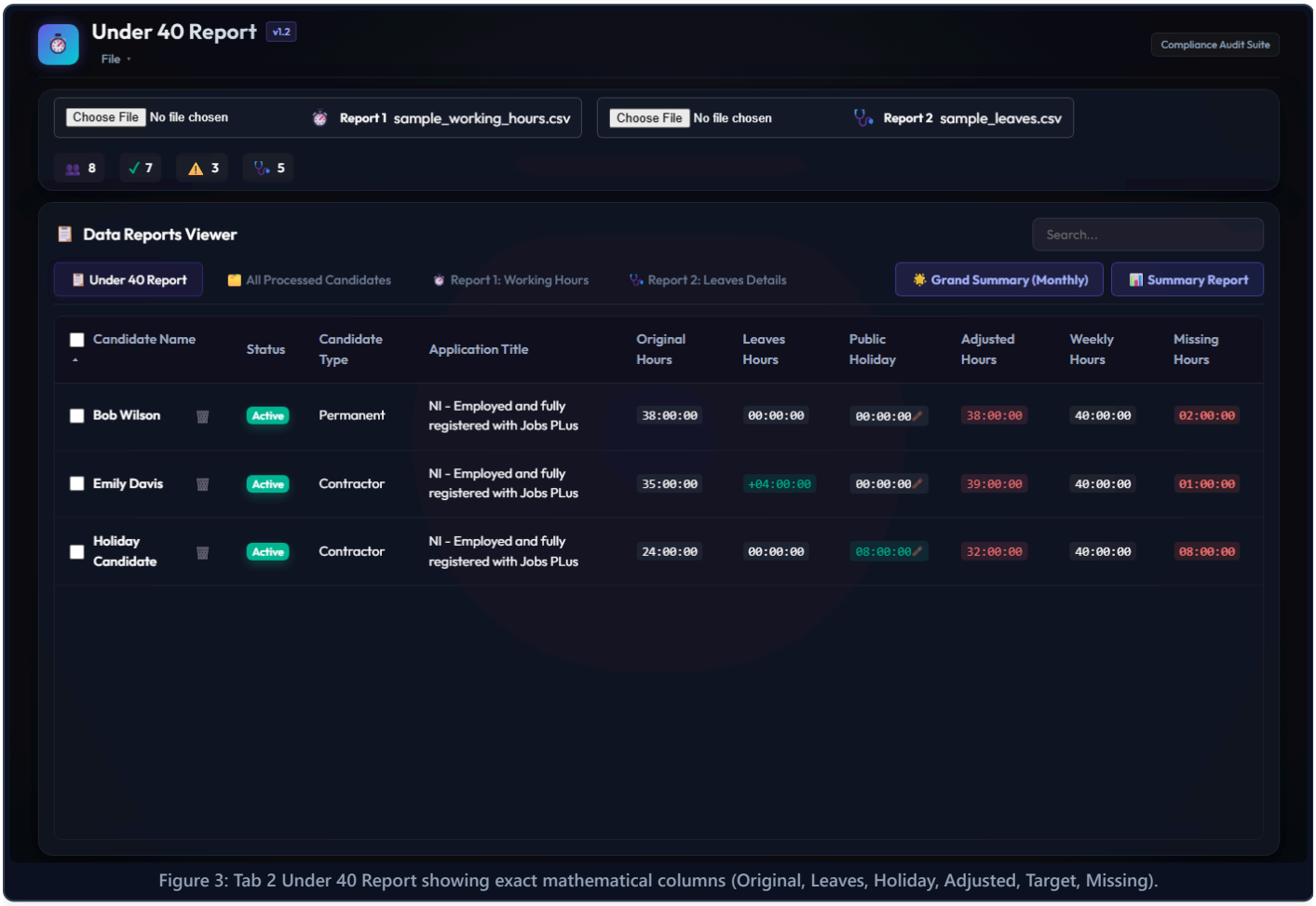


Figure 3: Tab 2 Under 40 Report showing exact mathematical columns (Original, Leaves, Holiday, Adjusted, Target, Missing).

Mathematical Columns & Metrics Displayed:

- **Candidate Name & Status:** Contractor identity and current employment status (e.g. *Active*).
- **Original Hours:** Unadjusted hours worked directly from timesheet clock-ins (e.g. `35:00:00`).
- **Leaves Hours:** Total approved medical appointments, sick leaves, and vacation hours added (e.g. `+04:00:00`).
- **Public Holiday:** Editable input allowing managers to credit statutory company holidays (e.g. `08:00:00`).
- **Adjusted Hours:** Verified mathematical sum of `Worked Hours + Leaves Hours + Public Holiday`.
- **Weekly Hours:** Standard contractual baseline requirement (`40:00:00`).
- **Missing Hours:** The exact compliance deficit highlighted in bold red (e.g. `-01:00:00`).

> **Complete Audit Traceability:** Every figure is mathematically verifiable. If an invoice or payroll deduction is questioned by a client or contractor, compliance officers can produce this exact screen or export to prove calculation accuracy.

9. Tab 3 Walkthrough: 🗂️ All Processed Candidates (Master Workforce Roster)

Tab 3 (All Processed Candidates) gives executive leadership the complete 360° view of the entire organization. It displays all active contractors—both those who met compliance and those who fell short:

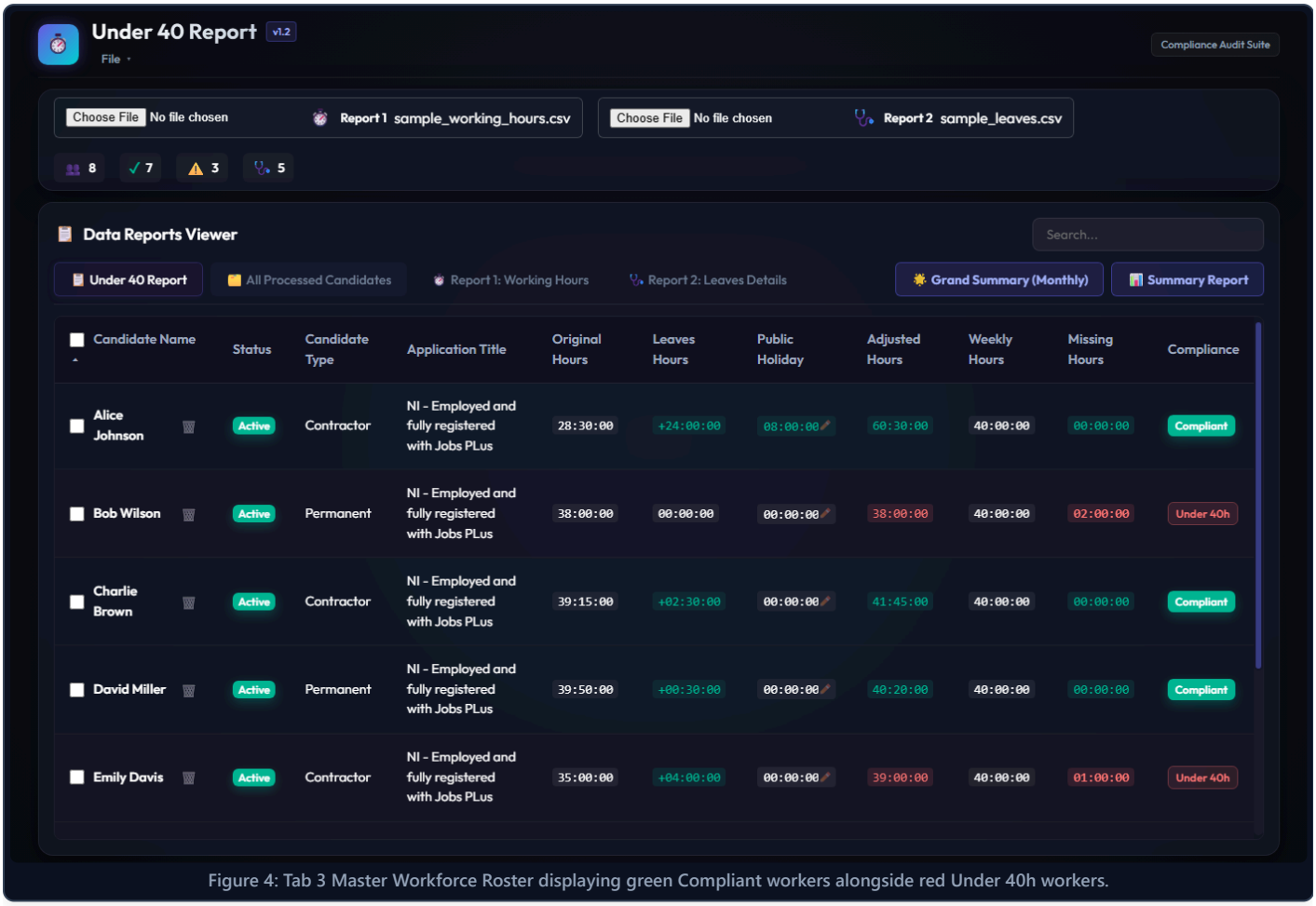


Figure 4: Tab 3 Master Workforce Roster displaying green Compliant workers alongside red Under 40h workers.

Key Features of the Master Roster:

🟢 Compliant Contractors

Candidates who reached 40+ hours are highlighted in green with a **Compliant** badge (e.g. Alice Johnson with 60.5h or Charlie Brown with 41.75h).

🔴 Under 40 Contractors

Candidates who fell short are clearly marked with a red **Under 40h** badge, making workforce exceptions impossible to miss.

🧹 Automatic Data Sanitization

Filters out dummy/test rows (e.g. "*TEST Peter*"), non-billable titles, and zero-hour placeholders, ensuring a pristine executive view.

⚡ Search As You Type

Search any contractor name or role instantly. Even across hundreds of workers, results filter in real-time with zero latency.

10. Tab 5 Walkthrough: 🌟 Grand Summary (Monthly Consolidation)

In monthly billing and invoicing cycles, managers must consolidate weekly shortfalls over a full month. The **Grand Summary** feature eliminates manual monthly cross-sheet reconciliations:

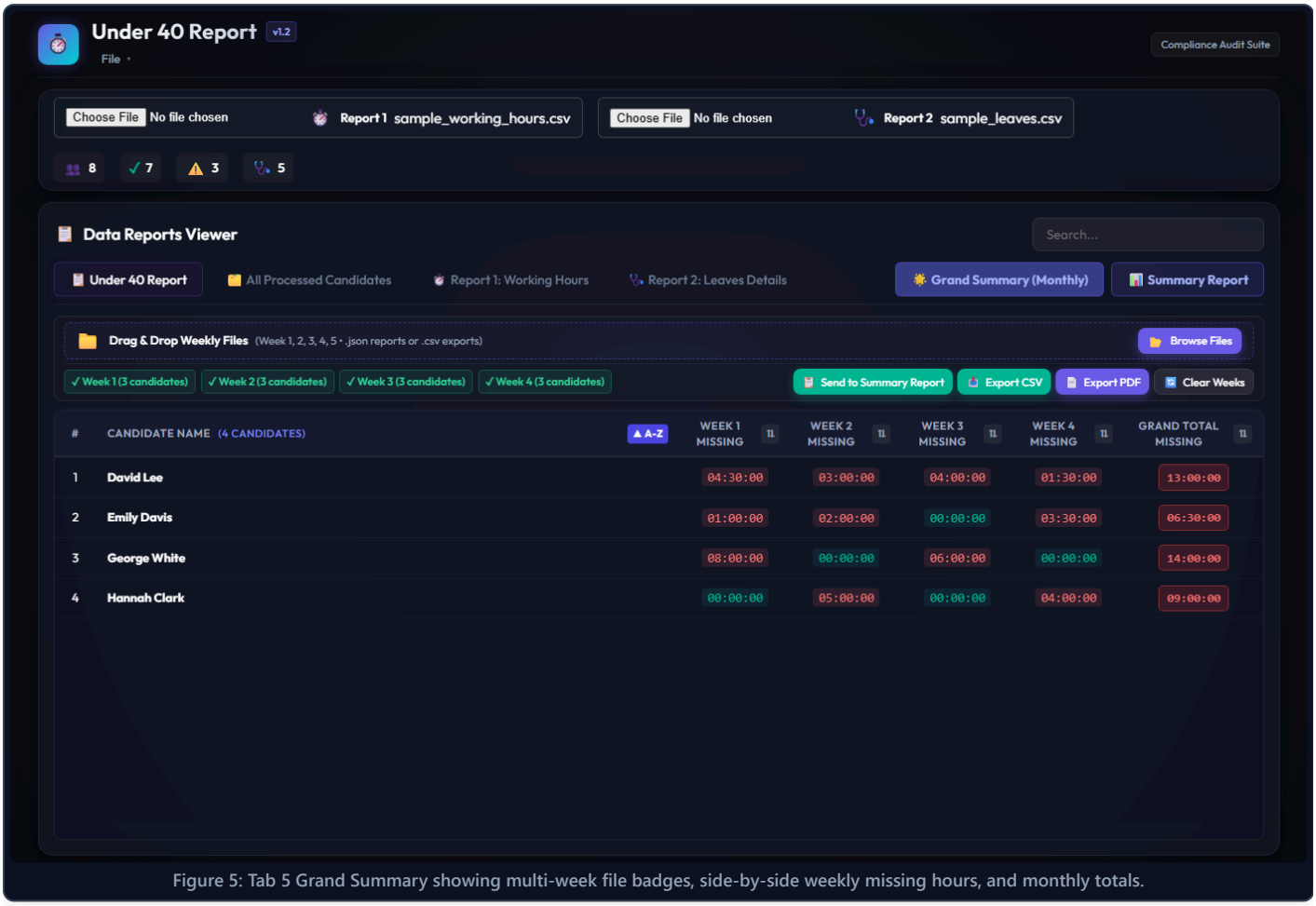


Figure 5: Tab 5 Grand Summary showing multi-week file badges, side-by-side weekly missing hours, and monthly totals.

How Monthly Consolidation Works:

1. Multi-Week Ingestion
Drag and drop all 4 or 5 weekly report files (`.json` audit sessions or `.csv` exports) onto the dedicated drop zone in one simple action.

2. Dynamic Week Badges
The dashboard renders visual status pills for each loaded week (e.g. *Week 1: 3 candidates, Week 2: 3 candidates...*), confirming full ingestion.

3. Multi-Week Deficit Aggregation
Automatically maps candidates across weeks using smart identity matching, displaying weekly shortfalls in dedicated columns alongside total monthly missing hours.

4. Send to Summary Report
One-click **"Send to Summary Report"** button transfers all monthly shortfall candidates into Tab 1 to assign Client Fault, Form of Pay, and Notes.

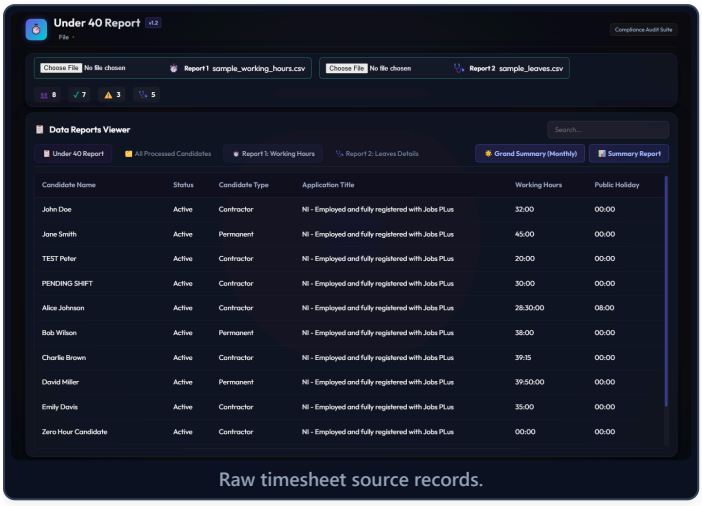
Sorting, Filtering & Monthly Reset

Managers can sort candidates alphabetically (**A-Z, Z-A**) or by **Total Missing Hours** to instantly identify chronically absent contractors. When the monthly cycle is complete, the **"Clear Weeks"** button resets the monthly view for the next period.

11. Raw Data Transparency, Report Chooser & Distribution

The application pairs unedited source data transparency with an interactive export chooser and portable desktop packaging:

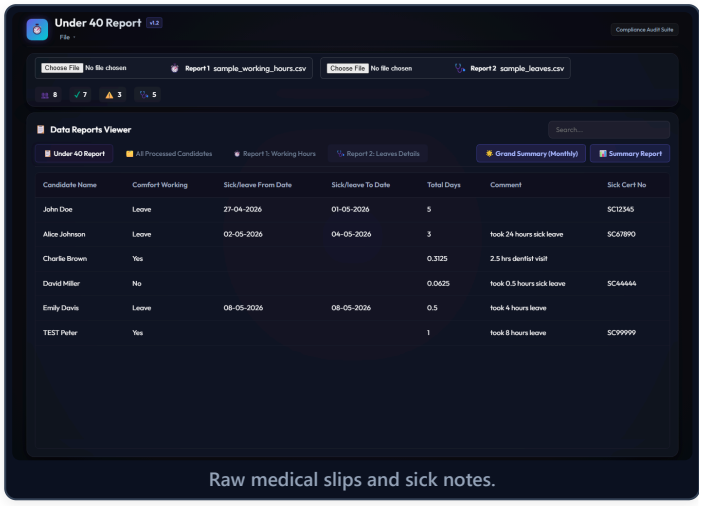
⏱️ Tab 3: Working Hours (Raw Source)



Raw timesheet source records.

Unedited timecards exactly as exported from the scheduling system. Allows instant column sorting and shift verification.

🩺 Tab 4: Leaves Details (Raw Source)



Raw medical slips and sick notes.

Doctor certificate numbers, leave dates, and original unedited comments, ensuring no sick slips were overlooked.

12. Interactive Export Chooser & White Paper Engine

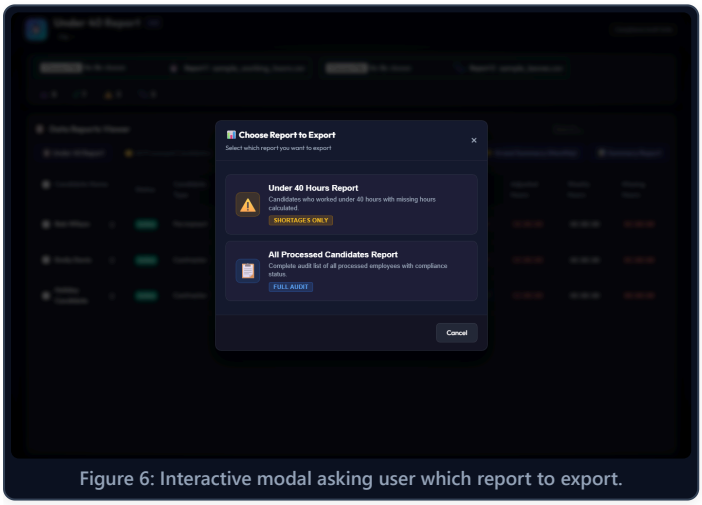


Figure 6: Interactive modal asking user which report to export.

Report Chooser Modal

When clicking Export CSV or Export PDF in the Data Reports Viewer, an intelligent dialog prompts the user:

- **Under 40 Hours Report:** Exports only candidates with hour shortfalls.
- **All Processed Candidates:** Exports complete roster with compliance statuses.

Clean White Paper PDF Reports

PDF exports feature an executive white paper layout with high contrast and zero colored background graphics to maximize printer efficiency.

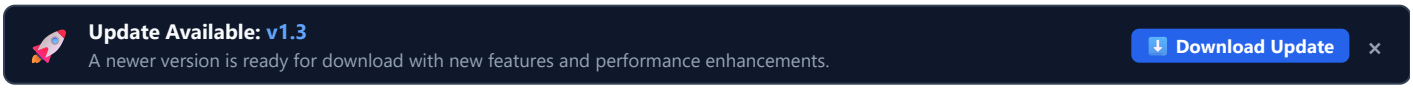
13. Portable Package & Automated In-App Updates

🚀 Portable Single-Click Executable

Ships as `Under40Report.exe` for Windows. Zero installation or admin rights required. Also includes `start_mac.command` for Mac/Linux.

🔔 Automated In-App Notifications

Whenever an update is released for the application, an in-app banner appears with a 1-click download link. You can also check under *"File > Check for Updates..."*.



14. Standard Operating Procedures (SOP) & Operational Cadence

To ensure consistent operational execution across teams, dispatch leads and managers should follow this standardized cadence:

Weekly Monday Morning Audit (5-Minute Cadence)

1. Launch `Under40Report.exe`.
2. Drag & drop **Report 1** (Working Hours) and **Report 2** (Leaves).
3. Verify candidate stats in top bar (Processed, Compliant, Under 40).
4. In Tab 1, select part-time workers and click "*Delete Part-Timer(s)*".
5. Assign **Dispatch Fault** and **Form of Pay** for all shortfalls.
6. Click "*Export PDF*" to generate the weekly management audit sheet.
7. Click "*File > Save Report (.json)*" to archive the session.

Monthly Billing & Consolidation Cadence

1. Navigate to **Tab 5: Grand Summary**.
2. Drag & drop all 4 or 5 weekly `.json` audit files.
3. Review cumulative candidate deficits across the billing month.
4. Sort by *Total Missing Hours* to identify chronic absenteeism.
5. Click "*Send to Summary Report*" to transfer monthly totals.
6. Complete billing adjustments and export the monthly PDF audit.
7. Click "*Clear Weeks*" to reset for the next monthly billing cycle.

15. Common Operational Edge Cases & Resolution Protocol

Contracted Part-Timers

Never leave part-timers on the shortfall audit. Check their row and click "*Delete Part-Timer(s)*". Restore at any time from the restore drawer.

Unstructured Doctor Slips

The parser automatically credits partial hours ("*4 hrs leave*") and defaults full-day slips to 8h. Click candidate row to override if required.

Statutory Public Holidays

When a bank holiday occurs, input the hours into the *Public Holiday* column in Tab 2 or drawer. Totals recalculate instantly.

16. Operational Benefits & Executive Summary

🎯 Zero Human Calculation Errors

Standardizes shortfall arithmetic, guaranteeing mathematical consistency across teams and ending payroll disputes before they reach leadership.

⚡ 100+ Hours Saved Every Year

Eliminates 2–3 hours of weekly spreadsheet drudgery, freeing operations staff to focus on high-value client communication and dispatch scheduling.

🔒 Enterprise Confidentiality & Code Shielding

100% client-side memory execution with scrambled code obfuscation and disabled developer tools ensures company logic, candidate records, and compensation numbers remain strictly secure and tamper-proof.

📦 Zero-Friction Desktop Tool

Single-click portable executable operates cleanly without installation, server setup, or admin rights, working completely offline or online.

> **Continuous Updates Guarantee:** You will always receive the latest updates whenever new features, compliance rules, or system improvements are released. The application will notify you directly with a 1-click download option, ensuring your operations and payroll workflows remain up to date at all times.